

Audio for every language: three engines, rendered once

Ochorus should **pre-render the whole library as audio, once**, and serve it as plain files. No single text-to-speech system covers all ten of our languages under a licence that lets us publish the audio, so we use three: **Kokoro** (open, self-hosted) for English, Spanish, French, Portuguese and Hindi; **Azure Neural TTS** (paid, one-time) for Arabic, Ukrainian, Swahili and Amharic; and **Sunbird AI** (open, Ugandan) for Luganda. Estimated total: **about \$650–850, paid once**, then about \$1 a month for storage.

1. Strategy

Speechify is ruled out: its terms do not allow audio generated for public distribution without a separate licence. That leaves three tests every option must pass, in this order:

- **Language.** All ten languages must be covered. Luganda is the binding constraint: no commercial service we found supports it.
- **Licence.** We must be allowed to store the audio and serve it to anyone. This rules out Meta MMS-TTS (CC-BY-NC), Coqui XTTS (non-commercial), Speechify, and Piper's Swahili and Arabic voices.
- **Cost.** No bill that grows with readers. Rendering once and serving files makes the cost a fixed, one-time amount tied to the size of the library, not to how many people listen.

Pre-rendering on a server also replaces the riskiest part of the earlier plan (*tts-plan.md*): running Kokoro inside the reader's browser, with an 80–160 MB download, iPhone memory limits and a device test matrix. A pre-rendered MP3 plays instantly on any phone, offline, and every chapter of a book sounds the same, as an audiobook should.

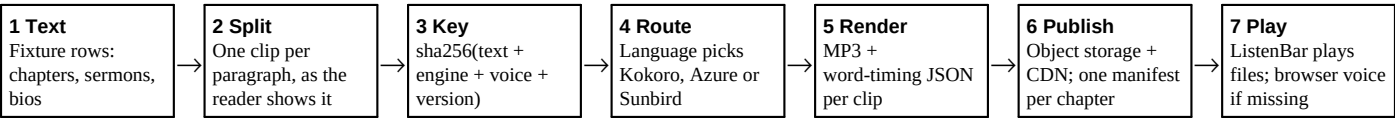
2. Which engine reads which language

Language	Text (M chars)	Engine	Why this engine
English	55.5	Kokoro	Best free quality; most of the library
Spanish · French	13.5 · 14.9	Kokoro	Supported natively; no cost
Portuguese	13.2	Kokoro	Brazilian voices; use Azure pt-PT if our text is European
Hindi	10.5	Kokoro	Supported; move to Azure if the listening test is weak
Swahili	21.6	Azure	Kenya + Tanzania voices; compare against Sunbird
Arabic · Ukrainian	7.0 · 5.9	Azure	Not in Kokoro; mature neural voices
Amharic	3.7	Azure	Only commercial engine we found with it
Luganda	6.5	Sunbird	Only usable option; needs native review
Total	152.3		~107 Kokoro · ~38 Azure · ~6.5 Sunbird

3. The three technologies

KOKORO-82M · open model		AZURE NEURAL TTS · cloud API		SUNBIRD AI · open model	
What	82M-parameter neural TTS; 54 voices in 8 languages.	What	Microsoft's neural voices; 140+ locales incl. sw-KE, sw-TZ, am-ET, uk-UA, ar-*.	What	Orpheus-3B fine-tune; 20 African languages incl. Luganda and Swahili.
Licence	Apache 2.0 weights. Audio is ours to publish.	Licence	Paid use; confirm terms allow storing and serving output publicly.	Licence	Apache 2.0, inheriting Meta's Llama 3 community licence.
Run it	Batch job on a rented GPU via Kokoro-FastAPI (Docker, OpenAI-style API).	Run it	Batch synthesis API from a Django management command; SSML for pauses.	Run it	Self-hosted on a rented GPU; slower than Kokoro (3B params).
Timings	Word timestamps from the same call.	Timings	Word-boundary events give word timings.	Timings	No word timings: align afterwards, or highlight per paragraph.
Watch	Uses espeak-ng (GPL) for phonemes: fine for a batch tool we do not ship.	Watch	~\$16 per 1M chars (HD ~\$22). 500K free/month.	Watch	Young model; a Luganda speaker must sign off.

4. Approach: how the audio is made and played



- **One clip per paragraph.** It matches today's Listen mode (*listen.svelte.ts*): the same highlight, skip and resume, now with word-by-word highlighting where the engine gives timings.
- **Content-addressed files.** Each clip is named by a hash of its text, engine, voice and engine version. A corrected paragraph gets a new hash, so only that clip is re-rendered and paid for; unchanged clips are never touched. Modern English and young-reader editions are separate rows, so they get their own audio with no special case.
- **One render command, three back ends.** A management command (e.g. *render_audio --book <slug> --lang <code>*) routes each language to its engine, skips clips that already exist, and writes a chapter manifest listing clip URLs, durations and timings. New works are rendered as part of publishing, like PDF/EPUB exports.
- **Plain files, cheap hosting.** ~2,800 hours of audio is about 40–60 GB of 32–48 kbps mono audio. Object storage with free egress (e.g. Cloudflare R2) costs about \$1 a month.
- **Graceful fallback.** If a manifest or clip is missing, ListenBar falls back to the browser's own voices, exactly as today. Nothing breaks while the library is part-rendered.
- **Fixed voice per language.** One house voice per language (and optionally a second) keeps a book consistent. Readers keep speed control; voice choice remains available through browser voices.

5. Cost (one-time, at list prices)

Engine	Text	How it is paid	Estimate
Kokoro	~107M chars (~2,000 h audio)	Rented GPU; runs far faster than real time	\$20–60
Azure Neural	~38M chars	\$16 per 1M characters	~\$610
Sunbird	~6.5M chars (~120 h audio)	Rented GPU; a 3B model, much slower	\$30–150
Storage + CDN	40–60 GB	Object storage, free egress	~\$1 / month
Total			~\$650–850 once

Contingencies: moving Hindi to Azure adds ~\$170; European Portuguese via Azure adds ~\$210. A single-vendor option (Azure for everything but Luganda) costs about \$2,300.

6. Rollout

Phase	What happens	Done when
0 Prove it	Render two passages per language on each candidate engine; native speakers rate them blind. Confirm Azure and Llama 3 licence terms.	An engine is chosen per language, in writing
1 Pilot	Build the render command, manifest and player support; render three flagship English books with Kokoro.	Pilot books play on iPhone, Android and desktop
2 Kokoro	Render all English, Spanish, French, Portuguese, Hindi.	~70% of the library has audio
3 Azure	Render Arabic, Ukrainian, Swahili, Amharic.	Nine of ten languages covered
4 Sunbird	Render Luganda; a native reviewer checks samples from every book.	All ten languages covered
5 Ongoing	New and corrected works render on publish.	No chapter without audio

7. Risks and how we handle them

Risk	Mitigation
A voice mispronounces names and Scripture	Pronunciation list per language (SSML / phoneme overrides); spot-check every book's first chapter
Licence terms change or are stricter than reported	Phase 0 gate; clips are tagged by engine, so one engine's audio can be re-rendered elsewhere
Luganda quality is not good enough	Ship Luganda on browser voices until a native reviewer approves
Audio drifts from corrected text	Content hashes: any text change re-renders that clip automatically

Sources (October 2026): Azure Speech language support and pricing; Kokoro-82M model card and Kokoro-FastAPI; Sunbird Orpheus-3B multilingual TTS model card and Sunflower v2 announcement; Speechify AI Voice API terms; Meta MMS-TTS and Piper voice model cards. Prices and licences were gathered by web search and must be confirmed with each vendor before purchase. Corpus sizes measured from the Ochorus content fixture.